Habiba Mohammed Mahfouz

CS Student @ Alexandria University | CCNA Certified & Network Enthusiast | Java & C++ Developer | Junior UI/UX

Alexandria, Egypt | WhatsApp | Gmail | GitHub | LinkedIn | Codeforces | Portfolio

PROFESSIONAL SUMMARY

Results-driven Software Industry & Multimedia (SIM) student at Alexandria University, specializing in bridging the gap between technical engineering and creative design. Expertly blending foundations in Secure Network Architecture (CCNA trained) and Software Development (Java, C++, Python) with professional UI/UX Product Design. Experienced in engineering scalable enterprise systems, 3D game logic, and data-driven multimedia campaigns.

TECHNICAL PROJECTS

Enterprise Network Architecture — Cisco Packet Tracer, CCNA

- Architected a secure and scalable enterprise network infrastructure using Cisco Packet Tracer.
- Implemented advanced topologies including VLAN segmentation, OSPF dynamic routing, and IPsec VPN tunneling.
- Applied stringent security controls (SSH, Port Security, DHCP Snooping) to mitigate network attacks.

Library Management System — C++

- Engineered a robust, console-based library automated system to manage book inventories and borrower logs.
- Leveraged core Object-Oriented Programming (OOP) principles to ensure a modular and maintainable code structure.
- Optimized system efficiency through structured data handling and complex algorithmic logic formulation.

Expression Evaluator — Java, Data Structures

- Developed a high-precision algorithmic expression calculator in Java using advanced data structures.
- Utilized Stacks and Linked Lists to parse mathematical infix operations and evaluate them via postfix conversion.
- Ensured system accuracy by handling complex parentheses and operator precedence through rigid algorithmic constraints.

Interactive Snake Game — Python

- Programmed a logic-driven Snake game featuring real-time input handling and state manipulation.
- Engineered fundamental game mechanics including rigid boundary constraints and self-collision algorithms.
- Implemented a dynamic continuous scoring system to ensure a fluid and engaging user experience.

Booki Store — Product Design (UI/UX)

- Directed an end-to-end UI/UX case study for an innovative mobile application that transforms reading into a social experience.
- Spearheaded data-driven user research (Personas, User Journeys) and delivered a high-fidelity interactive Figma prototype.
- Designed information architecture and gamified progress tracking modules to enhance user retention.

TECHNICAL SKILLS

Programming Core: Java, C++, Python (OOP, Data Structures, Algorithms).

Networking : CCNA (Routing & Switching), VLAN, OSPF, DHCP, VPN Tunneling, Network Security Fundamentals

Systems & Security: Windows Server (MCSA in progress), Kali Linux, Penetration Testing Basics

UI/UX design: UX Research, Information Architecture (IA), Wireframing, Prototyping

Creative Design: Adobe Illustrator, Audacity, Unity 3D.

Front-End Development: HTML, CSS, JavaScript

Tools: VS Code, Cisco Packet Tracer, Git/GitHub, NetBeans, Power BI.

Certifications & Education

- **B.Sc. in Software Industry & Multimedia (SIM) –** Faculty of science -Alexandria University (Expected 2028).
- **CCNA Routing and Switching –** Black Horse Alexandria (2025)
- **Software Fundamentals with C++ –** Code Spark (2025)
- **Introduction to Artificial Intelligence –** MCIT Egypt (2025)
- **Python & AI Development –** Black Horse Alexandria (2025)
- **UI & UX design** - Black Horse Alexandria (2026)

SOFT SKILLS

Problem Solving, Teamwork (Detailed team projects in portfolio), Self-Learning, Time Management

Languages

Arabic (Native), English, French